



DPUMixAmplifier

Additive mixing of two amplified input signals.

1. Inputs:

<i>In1</i>	Input 1 of mixing amplifier
<i>Multiplicator1</i>	Multiplicator of 1st input. Default: 1
<i>Divisor1</i>	Divisor of 1st input. Default: 1
<i>Offset1</i>	Offset of 1st input. Default: 0

<i>In2</i>	Input 2 of mixing amplifier
<i>Multiplicator2</i>	Multiplicator of 2nd input. Default: 1
<i>Divisor2</i>	Divisor of of 2nd input. Default: 1
<i>Offset2</i>	Offset of of 2nd input. Default: 0

2. Dynamic parameters:

<i>Gain</i>	Total gain. Default: 1
<i>Offset</i>	Offset of output. Default: 0
<i>Threshold</i>	Threshold value around neutral point of output. Default: 0
<i>MinOut, OutMin</i>	Minimum limit of output. Default: -1^{300} (no limit)
<i>MaxOut, OutMax</i>	Maximum limit of output. Default: 1^{300} (no limit)

3. Outputs:

<i>Out</i>	Output of mixing amplifier
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4. Functional principle:

The mixing amplifier adds both inputs *In1* and *In2*. Each of these inputs is amplified and has an offset added. Additionally the result of the addition is amplified and shifted by an offset:

$$Out = \left(\left(\frac{In_1 \cdot Multiplicator_1}{Divisor_1} \right) + Offset_1 \right) + \left(\left(\frac{In_2 \cdot Multiplicator_2}{Divisor_2} \right) + Offset_2 \right) \cdot Gain + Offset$$

If the absolute value of the output signal is smaller than *Threshold*, the output is set to 0.

$$\text{IF } |Out| < Threshold \text{ THEN } Out = 0$$

The range of the output is limited by both parameters *MinOut* and *MaxOut*.